E-Commerce Customer Conversion Prediction

Methodology & Performance Summary | Final F1-Score: **0.5626** | Decision Threshold: **0.24**

Problem & Metric

The objective of this challenge is to predict customer conversion (the binary target Converted) on an e-commerce platform using historical and session-based features. Submissions are evaluated using the **F1-Score** metric. The dataset has a mild class imbalance with a 30.87% positive conversion rate, which requires tuning the classification threshold rather than defaulting to 0.50 to maximize performance.

Exploratory Data Analysis (EDA)

EDA showed that the most critical signals for conversion are browsing behaviors: pages viewed (Pages_Viewed) and products viewed (Products_Viewed) both exhibit a strong correlation (r=0.31) with the target. Demographics (Age) and ID variables (Campaign_Code, Browser_Version) showed near-zero linear correlation. Critically, missing values in Income and Time_On_Site were found to be highly informative of low engagement, correlating with a lower conversion rate.

Data Preprocessing & Feature Engineering

To capture user behavior patterns cleanly, we implemented the following steps:

- **Ratio Features**: Computed browsing intensity ratios like pages viewed per product (Pages_Per_Product) and session duration per page/product (Time_Per_Page, Time_Per_Product).
- **Engagement Score**: Created a composite activity metric (Pages_Viewed * Products_Viewed * Session_Duration) representing session depth.
- **Interactions**: Multiplied income by age (Income_Age_Prod) and computed the income-to-age ratio (Income_Age_Ratio) to capture demographic purchasing power, alongside discount interactions (Discount_Purchases, Discount_Income).
- **High-Cardinality Handling**: Frequency-encoded the high-cardinality Campaign_Code (6,014 unique values) using the full population count, capturing campaign size while preventing target leakage.
- **Missing Value Flags**: Added binary missingness indicators (Age_missing, Income_missing, Time_On_Site_missing) to preserve the informative missingness signal before applying median imputations.

Model Selection & Cross-Validation

We selected tree-based ensembles (CatBoost Classifiers) for their ability to learn non-linear relationships, handle missing values, and natively manage categorical columns. We combined the training and public validation sets to train on 13,000 samples. A Stratified 5-Fold Cross-Validation scheme was used to evaluate models without leakage. The final prediction probabilities represent the average of the 5 CatBoost models.

Evaluation Results

The decision threshold was optimized on the out-of-fold probability predictions to maximize the F1-Score. The optimal threshold was found at **0.24**, which yielded a validation F1-Score of **0.5626** (a 43.4% relative improvement over the default 0.50 threshold F1 of 0.3923). The final predictions on the private test dataset predict **1668** conversions and **1332** non-conversions.

Model Family (5-Fold CV)	Decision Threshold	Validation F1-Score	Precision	Recall	Accuracy
Baseline Logistic Regression	0.50	0.3923	0.5050	0.3210	0.6975
LightGBM Classifier	0.27	0.5550	0.4421	0.7453	0.6347
CatBoost Classifier (FINAL)	**0.24**	**0.5626**	**0.4320**	**0.8064**	**0.6168**